DATA ANALYTICS PORTFOLIO REPORT

E-Commerce Sales & Customer Analytics

2021–2025

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MySQL · Python · pandas · Tableau Public

1. Executive Summary

E-Commerce Sales & Customer Analytics | 2021–2025

This project evaluates e-commerce sales performance, customer purchasing behavior, and product profitability using transactional data from January 2021 through December 2025. The objective is to identify revenue trends, understand customer purchasing patterns, and uncover opportunities to improve customer engagement and product performance.

Using MySQL, Python, and Tableau Public, the analysis combines three datasets covering sales transactions, order items, and product information. Two interactive dashboards were developed to monitor key performance indicators (KPIs) and explore sales, customer, and product trends.

Key Findings

• Sales Performance: The business generated $83.34 million in revenue and $32.86 million in profit across 67,978 completed USD orders, with an average order value of $1,226.05.

• Customer Behavior: Loyal customers contributed 93.94% of total revenue, highlighting the importance of established customer relationships and opportunities to strengthen new customer acquisition.

• Category Performance: Electronics was the highest-revenue product category, generating $19.34 million, while Grocery achieved the highest profit margin at 52.65%.

• Product Profitability: Product-level analysis revealed variations in revenue and profit margins, demonstrating the importance of evaluating both metrics when making merchandising and inventory decisions.

Business Implications

The findings suggest opportunities to strengthen customer acquisition, maintain engagement among loyal customers, and improve product strategies by balancing revenue generation with profitability. The analysis provides a foundation for data-driven decision-making through validated metrics, customer segmentation, and interactive performance monitoring.

2. Business Objectives & Questions

Project Objective

The primary objective of this project is to evaluate e-commerce business performance by analyzing sales transactions, customer purchasing behavior, and product profitability. The analysis aims to identify patterns and opportunities that can support informed decisions related to revenue growth, customer engagement, and product strategy.

Key Business Questions

Sales Performance

• How have revenue, profit, and order volume changed over time?

• Which periods generated the highest sales performance?

• What do average order values reveal about purchasing activity?

Customer Behavior

• How do new, returning, and loyal customers contribute to overall revenue?

• What patterns exist in customer spending and purchasing frequency?

• How does customer purchasing behavior vary across regions and time periods?

Product & Category Performance

• Which product categories and individual products generate the most revenue?

• Which categories achieve the highest profit margins?

• How does product profitability compare with revenue performance?

Business Opportunities

• Where are the greatest opportunities to improve customer acquisition and retention?

• How can revenue and profit margin insights inform product and inventory strategies?

• Which performance indicators should be monitored to support future business decisions?

Analytical Approach

The project uses SQL and Python to examine and validate transactional data, followed by Tableau Public to visualize performance metrics and explore business trends. Two interactive dashboards support the analysis: Executive Overview, which focuses on overall business performance, and Customer & Product Analysis, which examines customer segments, purchasing patterns, and product profitability.

3. Dataset Overview

Data Sources

The analysis uses three interconnected e-commerce datasets containing sales transactions, individual order items, and product information. Together, these datasets provide the foundation for evaluating business performance, customer purchasing behavior, and product profitability.

Dataset	Records	Description
Sales Transactions	138,116	Order-level information, including transaction dates, customer details, sales amounts, profit, and order status.
Order Items	397,569	Individual products purchased within each order, including quantities, revenue, costs, and profit.
Product Catalog	1,175	Product information, including categories, subcategories, brands, and pricing.

Data Relationships

The datasets are connected using two primary identifiers:

• Order ID: Connects sales transactions to individual order items.

• Product ID: Connects order items to product information.

These relationships allow the analysis to examine overall business performance while also exploring revenue and profitability at the product and category levels.

Analytical Scope

The datasets cover transactions from January 2021 through December 2025. To maintain consistency in financial reporting, the analysis focuses on completed orders recorded in USD. After applying these restrictions, the order-level dataset contains 67,978 completed USD orders, representing 14,783 unique customers. The analysis evaluates sales performance, customer segments, purchasing patterns, and product profitability across the five-year period.

Data Considerations

Sales transactions and order items contain financial measures at different levels of detail. Because the source datasets do not fully reconcile financially, order-level sales data is used for overall revenue and profit KPIs, while item-level data supports product and category analysis. This distinction is maintained throughout the dashboards and report to avoid combining financial measures that are not directly comparable.

4. Data Preparation & Validation

Data Quality Assessment

Before conducting the analysis, the three datasets were examined using Python, pandas, and MySQL to assess data completeness, consistency, and relational integrity. The assessment included checking for missing values, duplicate records, inconsistent data types, and relationships between datasets.

• Missing Values: Missing values were primarily identified in the return_status and coupon_code fields. These fields were not required for the primary sales, customer, and product performance calculations.

• Duplicate Records: A small number of duplicate records were identified during the initial assessment. Duplicate checks were used to evaluate potential data-quality concerns.

• Referential Integrity: All order IDs in the Order Items dataset matched records in Sales Transactions, and all product IDs referenced in Order Items matched the Product Catalog.

• Currency Consistency: Transactions included multiple currencies. Analysis was restricted to USD transactions to avoid combining monetary values without exchange-rate conversion.

Financial Reconciliation

Financial totals were compared between the Sales Transactions and Order Items datasets to assess consistency across reporting levels. The reconciliation identified discrepancies in revenue, costs, and profit between the two datasets. For example, total gross sales were approximately $189.96 million in Sales Transactions compared with $206.34 million in Order Items before applying the final analytical filters.

Because these discrepancies could not be explained solely by differences in reporting granularity, the datasets were not treated as financially interchangeable. To maintain consistent reporting:

• Order-level data was used for overall revenue, profit, average order value, and customer financial metrics.

• Item-level data was used for product and category revenue and profitability analysis.

• Financial measures from the two sources were not combined into a single reconciled total.

Data Filtering & Preparation

The analysis focused on completed orders recorded in USD. These restrictions were applied consistently to the Tableau data sources. Sales and product information were connected through order and product identifiers, allowing analysis across different levels of detail without unnecessarily duplicating order-level records. Customer spending calculations were also configured to respond to relevant dashboard filters, supporting analysis by year, region, and customer type.

Validation Approach

SQL queries were used to verify key dashboard metrics, including total revenue, total profit, order counts, customer counts, category performance, and customer spending distributions. Results were compared against Tableau calculations to confirm consistency between the underlying data and the published visualizations. This process helped establish a reliable analytical foundation while documenting limitations in the source data.

5. Sales Performance Analysis

Overall Sales Performance

From January 2021 through December 2025, the business generated $83.34 million in revenue and $32.86 million in profit from 67,978 completed orders recorded in USD.

Key Performance Indicator	Value
Total Revenue	$83,344,126.98
Total Profit	$32,860,657.92
Total Orders	67,978
Total Customers	14,783
Average Order Value	$1,226.05

The business generated approximately $0.39 in profit for every dollar of reported net sales, indicating an overall profit-to-revenue ratio of approximately 39.4%. The average order value of $1,226.05 provides a baseline for evaluating transaction-level purchasing behavior and monitoring changes in customer spending over time.

Monthly Revenue Trends

Monthly revenue analysis identified December 2023 as the highest-revenue month, generating approximately $2.28 million. This result highlights a period of particularly strong sales performance. However, additional analysis would be required to determine whether the increase was driven by seasonal demand, promotional activity, higher order volume, or changes in average order value.

Monitoring monthly revenue alongside order counts and average order value can help distinguish between growth driven by transaction volume and growth driven by higher spending per transaction.

Revenue & Profitability Insights

Although revenue and profit are closely related, evaluating both measures provides a more complete understanding of business performance. Revenue measures the value of completed sales, while profit reflects the financial contribution remaining after the costs represented in the source data. The overall profit-to-revenue ratio provides a useful benchmark for tracking profitability over time and evaluating changes across customer segments and reporting periods.

Business Implications

• Monitor monthly revenue and order volume together to identify whether changes in sales performance are driven by purchasing frequency or transaction value.

• Investigate high-performing periods, particularly December 2023, to identify factors that may have contributed to stronger revenue.

• Track profitability alongside revenue to ensure that sales growth translates into favorable financial outcomes.

These findings establish a baseline for evaluating customer purchasing behavior and product performance in subsequent sections.

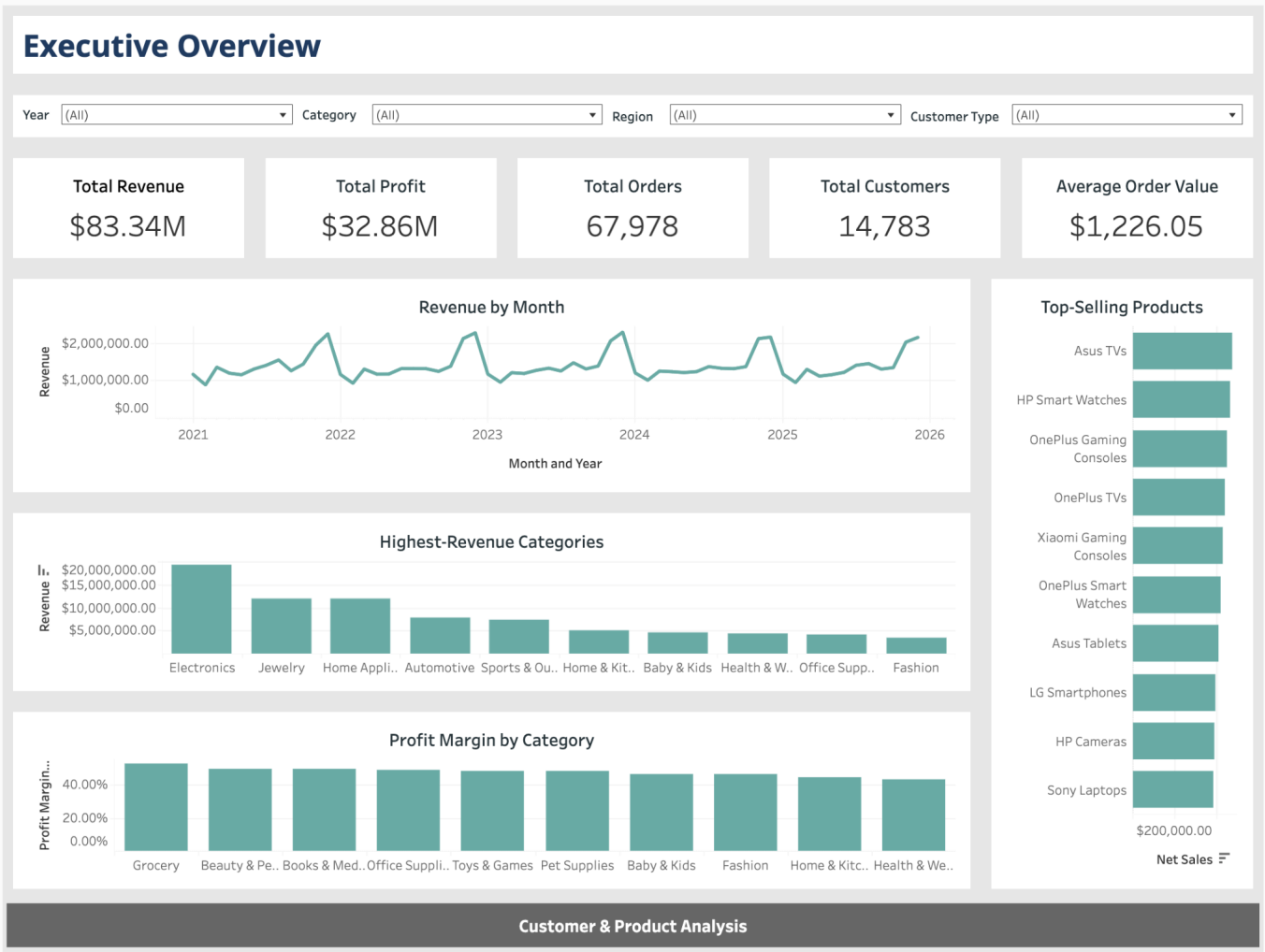


Figure 1. Executive Overview dashboard (all filters).

6. Customer Analysis

Customer Overview

The analysis identified 14,783 unique customers across 67,978 completed USD orders between January 2021 and December 2025. The average customer value was $5,637.84, indicating the average revenue generated per customer during the analysis period. Customer behavior was evaluated through revenue contributions, purchasing patterns, and spending distributions to understand differences among customer segments.

Revenue by Customer Type

Revenue was heavily concentrated among customers classified as Loyal.

Customer Type	Revenue	Share of Revenue
Loyal	$78,297,402.26	93.94%
Returning	$4,830,310.32	5.80%
New	$216,414.40	0.26%

Loyal customers accounted for the overwhelming majority of revenue, demonstrating the importance of established customer relationships to overall business performance. However, the limited contribution from New customers suggests an opportunity to investigate customer acquisition and early-stage engagement. Because these classifications come from the source dataset, additional analysis would be needed to determine how customers transition between segments over time.

Customer Spending Distribution

Customer-level spending analysis revealed variation in total purchasing value, with customers distributed across multiple spending ranges. The spending distribution provides insight into differences in customer purchasing activity and helps identify segments that contribute disproportionately to overall revenue. Examining these patterns alongside customer type and region can support more targeted marketing strategies and customer engagement initiatives.

Repeat Purchasing Behavior

Repeat purchasing behavior was examined using customer purchase history and the repeat purchase rate KPI. The analysis supports evaluating the role of repeat transactions in revenue generation and understanding how customer purchasing patterns vary across different segments. Monitoring repeat purchasing alongside customer acquisition can help businesses assess the balance between attracting new customers and maintaining relationships with existing customers.

Business Implications

• Strengthen customer acquisition: Investigate why New customers account for a relatively small share of revenue and evaluate opportunities to improve acquisition and conversion.

• Maintain loyal customer engagement: Monitor the purchasing behavior of Loyal customers and explore strategies that encourage continued engagement.

• Develop targeted customer strategies: Use customer spending distributions, purchase history, and regional patterns to identify opportunities for more personalized marketing and retention initiatives.

Overall, the findings highlight the importance of understanding customer segments and purchasing behavior when developing sustainable revenue strategies.

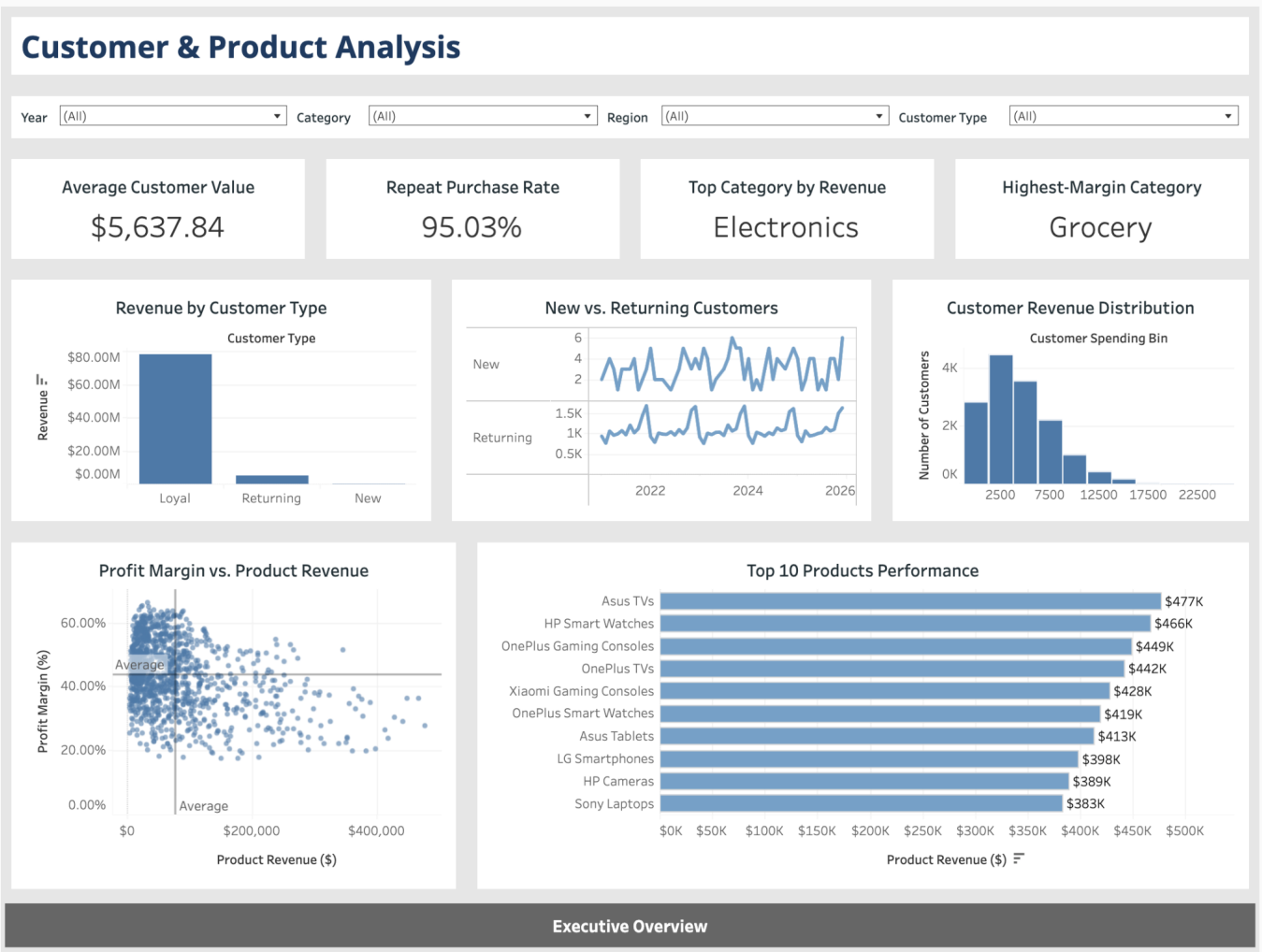


Figure 2. Customer & Product Analysis dashboard (all filters).

7. Product & Category Analysis

Category Revenue Performance

Product category analysis identified Electronics as the highest-revenue category, generating approximately $19.34 million in item-level revenue from completed USD orders. This result demonstrates the category's substantial contribution to product sales and highlights its importance when evaluating overall product performance. However, revenue alone does not provide a complete picture of financial performance. Comparing category revenue with profit margins helps identify differences between sales volume and profitability.

Category Profitability

While Electronics generated the highest revenue, Grocery achieved the highest profit margin at 52.65%. This distinction demonstrates that the categories generating the most revenue are not necessarily those delivering the highest percentage returns. Higher-margin categories may offer opportunities to improve profitability through merchandising, pricing, and promotional strategies. However, margin percentages should also be evaluated alongside total revenue and profit contributions.

Top-Performing Products

Product-level analysis identified differences in revenue generation across individual products. The highest-revenue product was an Asus television (PROD-000074), generating approximately $476,863 in item-level revenue. The Top 10 Products Performance visualization highlights products contributing the most revenue, while the Profit Margin vs. Product Revenue scatterplot compares revenue and profitability across the product catalog. Together, these visualizations help distinguish products with strong sales performance from those generating higher profit margins.

Revenue vs. Profit Margin

The relationship between product revenue and profit margin is important when evaluating product performance. Products generating substantial revenue may operate at relatively lower margins, while products with smaller revenue contributions may achieve higher percentage margins. Evaluating both measures provides a more balanced understanding of product performance than ranking products by revenue alone.

Business Implications

• Optimize product strategies: Evaluate high-revenue products alongside their profit margins to identify opportunities to improve profitability.

• Investigate high-margin categories: Examine categories such as Grocery to determine whether additional marketing or merchandising investment could support profitable growth.

• Improve inventory decisions: Use product revenue and profitability trends to inform assortment planning, inventory prioritization, and promotional strategies.

Overall, the findings demonstrate the importance of balancing revenue generation with profitability when making product-related business decisions.

8. Key Insights & Recommendations

The analysis identified several opportunities to improve customer engagement, strengthen revenue performance, and optimize product profitability. The following recommendations are based on the findings from the sales, customer, and product analyses.

1. Strengthen New Customer Acquisition

Key Insight: Loyal customers generated approximately 93.94% of total revenue, while New customers contributed only 0.26%.

Business Implication: Revenue is heavily concentrated among established customers. Although this demonstrates the importance of loyal customers, it also suggests a need to investigate the effectiveness of customer acquisition efforts.

Recommendation: Evaluate acquisition channels, marketing campaigns, and new customer conversion rates to identify opportunities for attracting and engaging new customers. Monitor new customer acquisition alongside repeat purchasing behavior to assess long-term customer growth.

2. Maintain and Develop Loyal Customer Relationships

Key Insight: Loyal customers generated approximately $78.30 million in revenue, making them the largest contributing customer segment.

Business Implication: Changes in loyal customer purchasing behavior could have a substantial impact on overall revenue.

Recommendation: Monitor purchasing frequency, customer spending, and repeat purchase rates. Consider targeted engagement strategies, personalized promotions, and loyalty initiatives to support continued purchasing activity.

3. Balance Product Revenue with Profitability

Key Insight: Electronics generated the highest category revenue at approximately $19.34 million, while Grocery achieved the highest profit margin at 52.65%.

Business Implication: Revenue leadership does not necessarily indicate the strongest percentage profitability.

Recommendation: Evaluate category and product performance using both revenue and profit margin metrics. Investigate whether high-margin categories offer opportunities for profitable growth while continuing to monitor the financial contributions of high-revenue categories.

4. Investigate High-Performing Sales Periods

Key Insight: December 2023 generated the highest monthly revenue, approximately $2.28 million.

Business Implication: Identifying the factors associated with stronger sales periods could help improve future planning and performance forecasting.

Recommendation: Compare high-performing months with marketing activity, order volume, average order value, and product demand. Use these findings to inform promotional timing, inventory planning, and sales forecasting.

5. Establish Consistent Performance Monitoring

Key Insight: The analysis demonstrates how revenue, profitability, customer behavior, and product performance provide complementary perspectives on business health.

Business Implication: Monitoring individual metrics in isolation may overlook important relationships between sales growth, customer engagement, and profitability.

Recommendation: Use the interactive Tableau dashboards to monitor KPIs regularly. Track revenue, profit, average order value, customer segments, and product margins together to support more informed business decisions.

Recommended Priorities

The most immediate priorities are to investigate new customer acquisition, maintain engagement among loyal customers, and evaluate product profitability alongside revenue. These recommendations represent opportunities for further investigation rather than guaranteed outcomes. Additional marketing, customer acquisition, and operational data would be needed to assess their effectiveness.

9. Limitations & Methodology

Analytical Scope

The analysis covers e-commerce transactions from January 2021 through December 2025 and focuses exclusively on completed orders recorded in USD. Transactions in other currencies were excluded to avoid combining financial values without applying exchange-rate conversions. Consequently, the findings represent a subset of the original dataset rather than the business’s complete transaction history.

Financial Data Limitations

Financial reconciliation identified differences between the Sales Transactions and Order Items datasets. Because these discrepancies could not be fully resolved using the available information, the analysis maintained separate reporting approaches:

• Order-level Sales Transactions: Used for overall revenue, profit, average order value, and customer financial metrics.

• Item-level Order Items: Used for product and category revenue and profitability analysis.

Financial totals from these sources should not be interpreted as directly interchangeable.

Customer Classification

Customer types—New, Returning, and Loyal—were based on classifications provided in the source data. These classifications were not independently reconstructed using each customer's chronological purchase history. Therefore, findings describe the revenue contributions of the recorded customer segments rather than verified customer lifecycle transitions.

Data Quality Considerations

The initial data assessment identified limited missing values and a small number of duplicate records. Missing values were concentrated in fields such as return_status and coupon_code, which were not central to the primary financial calculations. Data relationships and dashboard calculations were validated using SQL queries. However, the analysis remains dependent on the accuracy and completeness of the original source records.

Dashboard Filter Methodology

The interactive Tableau dashboards support analysis by Year, Region, Customer Type, and Product Category. Year, Region, and Customer Type filters allow users to examine performance across selected segments. The Product Category filter applies only to product- and category-related visualizations. It does not modify overall financial KPIs, which are calculated from order-level sales data. Customer spending calculations were configured to respond to relevant filters, allowing spending distributions to reflect the selected analytical scope.

Additional Limitations

The analysis is descriptive and does not establish causal relationships between business activities and financial outcomes. For example, high monthly revenue does not independently demonstrate the effectiveness of promotional campaigns, and a low revenue contribution from New customers does not necessarily indicate poor acquisition performance. Additional marketing, inventory, operational, and customer lifecycle data would be required to investigate these relationships further.

10. Conclusion

The E-Commerce Sales & Customer Analytics project provides a comprehensive assessment of sales performance, customer purchasing behavior, and product profitability from January 2021 through December 2025.

The analysis identified several important findings, including the significant revenue contribution of loyal customers, Electronics' leadership in category revenue, and Grocery's comparatively strong profit margins. These results highlight opportunities to strengthen customer acquisition, maintain customer engagement, and optimize product strategies by evaluating both revenue and profitability.

Through the use of MySQL, Python, pandas, and Tableau Public, the project demonstrates the application of data validation, relational data analysis, KPI development, and interactive data visualization. The resulting dashboards provide a framework for exploring business performance across different time periods, customer segments, and product categories.

The project also emphasizes the importance of data quality and transparent reporting. Financial discrepancies between source datasets were documented, and analytical limitations were considered when interpreting results and developing recommendations.

Overall, this analysis demonstrates how e-commerce transactional data can be transformed into meaningful business insights. By combining technical analysis with business-focused interpretation, the project illustrates how data analytics can support informed decision-making, performance monitoring, and strategic planning.